

Audio Volume Envelope Analysis

Segment-Level Statistical Report

July 13, 2026

1 Overview

This report analyzes the volume (loudness) envelope of an audio recording that was split into **5 segments**, each approximately **57.3 seconds** long, for a total analyzed duration of **286.5 seconds** (~4 minutes 46 seconds). For every segment, the source data provides a time series of instantaneous volume expressed as a percentage of the maximum observed level (0–100%), sampled at a rate of approximately **43.1 samples per second** (2,468 samples per segment). Each sample is additionally paired with a hex color code drawn from the **plasma** colormap, consistent with the color encoding typically used to render spectrogram intensity.

The purpose of this analysis is to summarize the loudness behavior of each segment, compare segments against one another, and highlight structural patterns such as silence, sustained loudness, and periodicity.

2 Summary Statistics

Table 1 reports descriptive statistics for the volume-percentage series of each segment: the mean, population standard deviation, median, and observed range.

Table 1: Descriptive statistics of volume (%) by segment

Segment	Duration (s)	Samples	Rate (Hz)	Mean	Std. Dev.	Median	Min–Max
1	57.30	2468	43.07	44.88	18.93	48.72	0.00–100.00
2	57.30	2468	43.07	47.87	18.23	49.07	6.00–100.00
3	57.30	2468	43.07	47.35	17.82	48.46	1.03–100.00
4	57.30	2468	43.07	28.67	24.98	26.41	0.62–100.00
5	57.30	2468	43.07	31.86	18.06	31.13	0.00–100.00
Overall	286.49	12340	43.07	40.13	21.42	–	0.00–100.00

Segments 1–3 behave as a coherent group, with mean loudness clustered tightly between 44.9% and 47.9% and comparable variability ($\sigma \approx 18$ –19). Segments 4 and 5 are markedly quieter, with segment 4 showing both the lowest mean (28.67%) and the highest variability ($\sigma = 24.98$), indicating a segment that alternates between extended quiet stretches and occasional loud peaks rather than sustaining a consistent level.

3 Volume Envelope Over Time

Figure 1 shows the full volume envelope for each segment plotted against time, with the segment mean overlaid as a dashed line. All five segments exhibit a clear **quasi-periodic structure**: the envelope rises sharply from near zero to a local plateau in the 50–60% range, sustains that plateau

with a slow decay over roughly 2–3 seconds, and then drops again before the cycle repeats. This repeating attack-sustain-decay pattern is characteristic of rhythmic or phrase-level structure in the source audio (e.g., recurring words, notes, or beats) rather than a continuous, unstructured signal.

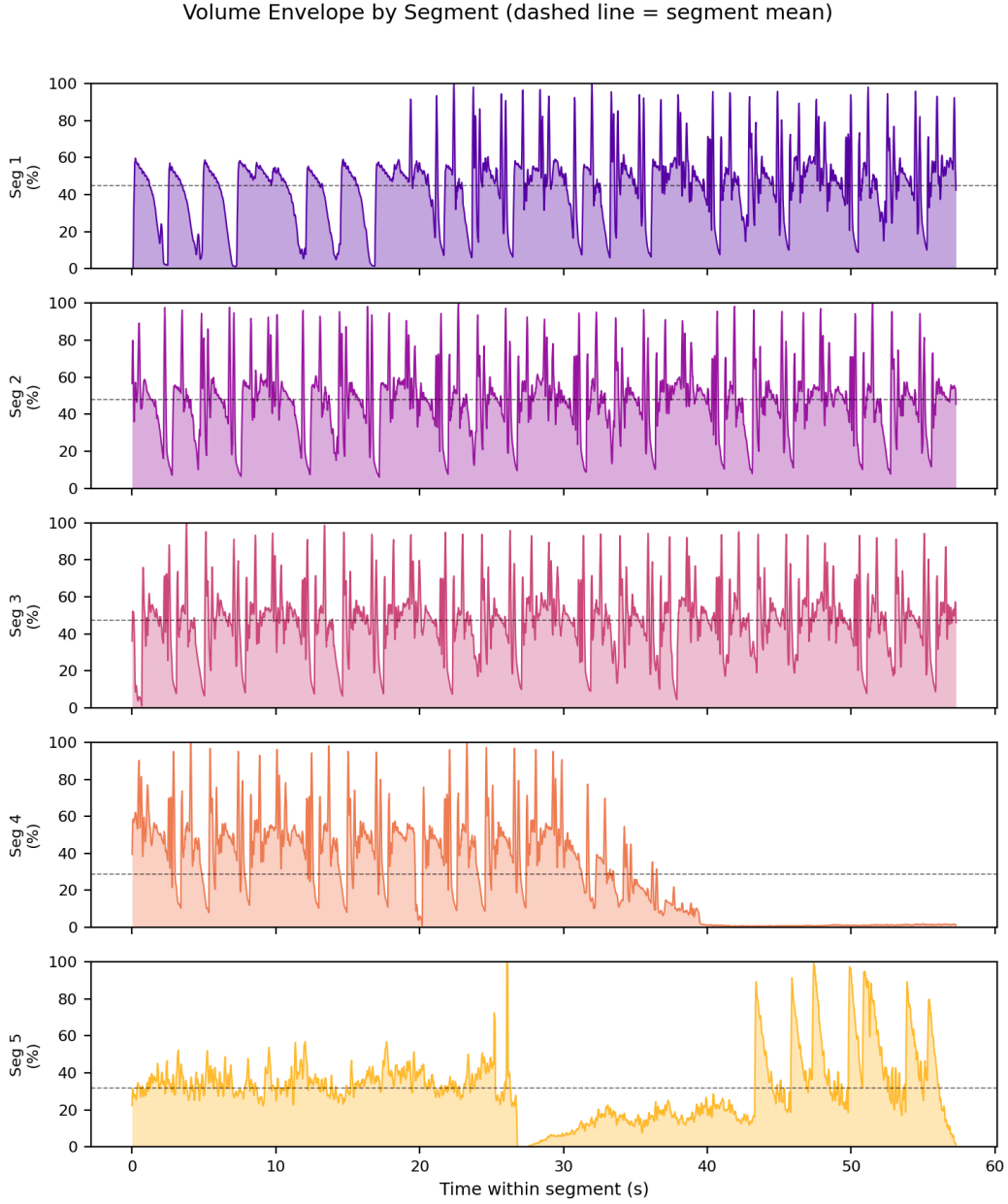


Figure 1: Volume envelope (%) versus time within each 57.3-second segment. The dashed horizontal line marks the mean level for that segment.

Segment 4 stands out visually: long flat stretches near 0% are interrupted by short bursts that reach 50–90%, consistent with the elevated standard deviation reported in Table 1 and the much lower mean.

4 Volume Band Composition

To characterize how each segment spends its time across the loudness range, samples were binned into five bands: *Silence* ($< 5\%$), *Quiet* ($5\text{--}30\%$), *Moderate* ($30\text{--}60\%$), *Loud* ($60\text{--}85\%$), and *Very Loud* ($\geq 85\%$). The resulting composition is shown in Table 2 and Figure 2.

Table 2: Percentage of time spent in each volume band, by segment

Segment	Silence ($<5\%$)	Quiet ($5\text{--}30\%$)	Moderate ($30\text{--}60\%$)	Loud ($60\text{--}85\%$)	Very Loud ($\geq 85\%$)
1	2.1%	18.4%	68.5%	8.3%	2.8%
2	0.0%	16.0%	67.5%	12.7%	3.9%
3	0.4%	16.0%	66.2%	14.5%	3.0%
4	31.5%	20.4%	39.1%	7.1%	1.9%
5	3.9%	42.9%	45.7%	5.6%	2.0%

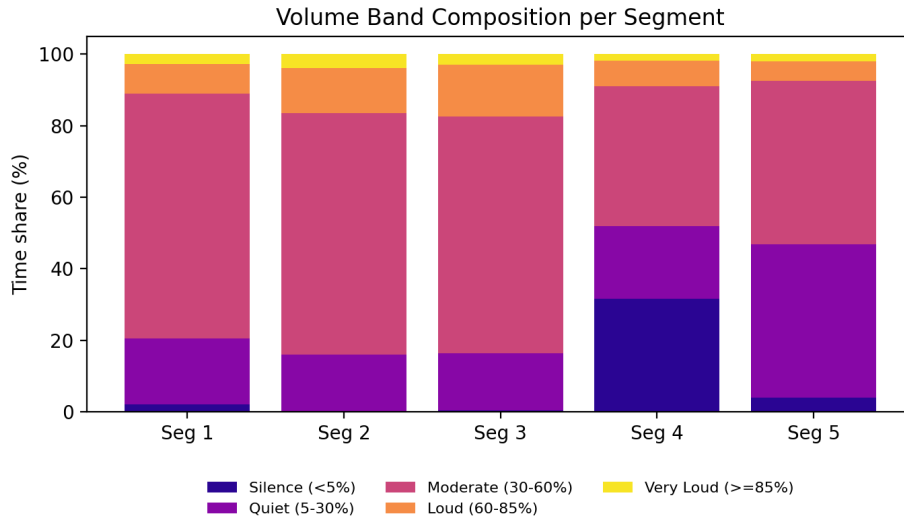


Figure 2: Stacked composition of time spent in each volume band, per segment.

Segments 1–3 are dominated by *Moderate* loudness (66–69% of samples), reflecting a sustained mid-level signal typical of continuous speech or music. Segment 4 is the clear outlier: **31.5% of its duration is silent**, roughly fifteen times the silence share of any other segment, which strongly suggests a pause, transition, or quiet passage located within this segment. Segment 5 leans toward the *Quiet* band (42.9%), indicating an overall softer passage without the extended silence seen in segment 4.

5 Distribution Shape and Segment Comparability

Figure 3 compares the full distribution of volume values across segments using box plots (outliers omitted for clarity), while Figure 4 shows the pooled distribution across all 12,340 samples.

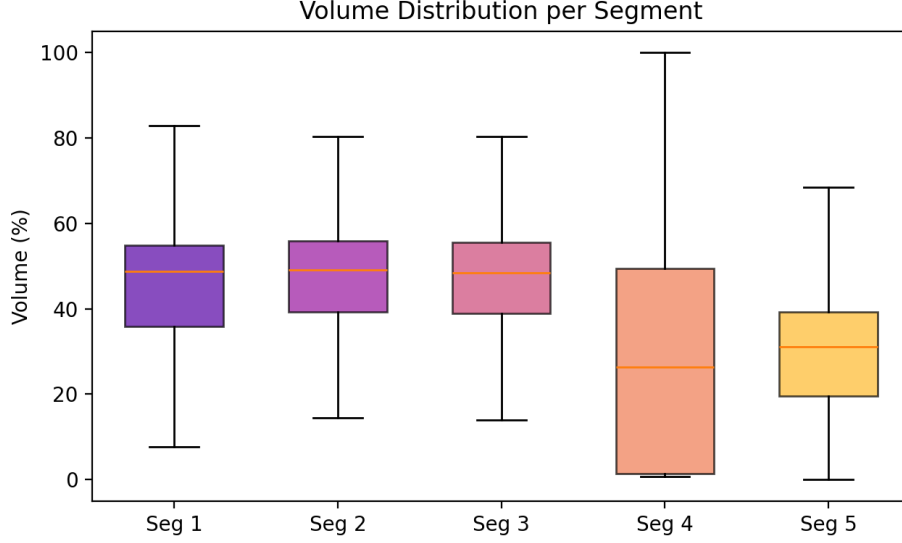


Figure 3: Distribution of volume (%) per segment.

The interquartile ranges for segments 1–3 sit at similar, elevated levels, confirming their similarity noted in Table 1. Segment 4’s box is shifted downward and compressed near the lower end, with its median (26.41%) well below the other segments, while its long whisker toward higher values reflects the intermittent loud bursts identified in Figure 1.

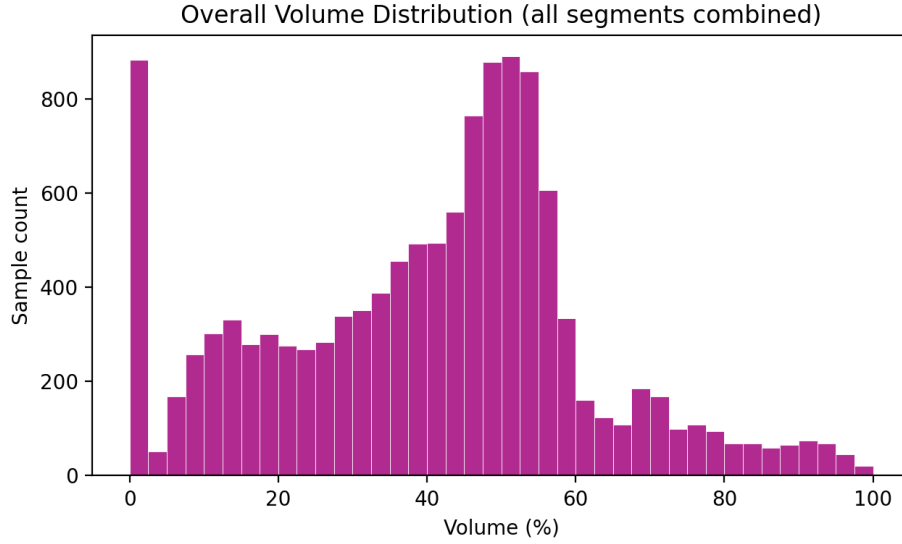


Figure 4: Pooled distribution of volume (%) across all segments and samples.

The pooled histogram is broadly unimodal with a peak in the 45–55% range, consistent with the moderate-band dominance seen across segments, and a long tail extending toward 0% driven primarily by segment 4’s silent stretches.

6 Key Findings

- The recording maintains a moderate average loudness overall (mean = 40.13%, $\sigma = 21.42$), with no segment ever reaching sustained near-silence or sustained maximum loudness.
- Segments 1, 2, and 3 form a homogeneous group: similar means (44.9–47.9%), similar variability, and near-identical band composition, suggesting continuous, evenly-paced content.
- Segment 4 is the clearest structural outlier, defined by extensive silence (31.5% of its duration) and the highest variability of any segment ($\sigma = 24.98$), pointing to a pause or quiet interlude followed by isolated loud events.
- Segment 5 is intermediate: quieter than segments 1–3 on average but without segment 4’s extended silence, instead spending most of its time in the *Quiet-to-Moderate* range.
- All segments display a repeating attack-decay envelope shape with a period on the order of 2–3 seconds, indicating rhythmic or phrase-based structure in the underlying audio rather than a flat or purely random loudness profile.